AI AND DATA VISUALISATION · ASSIGNMENT 2

The pattern that wasn’t there

Bias and ethics in a Tableau Pulse insight

Amadevborho Abusomwan · PgD Applied AI, ATU Donegal

Background

Section 1

Dataset

Retail Sales · 432 rows · 36 months
Jan 2023 – Dec 2025

AI tool

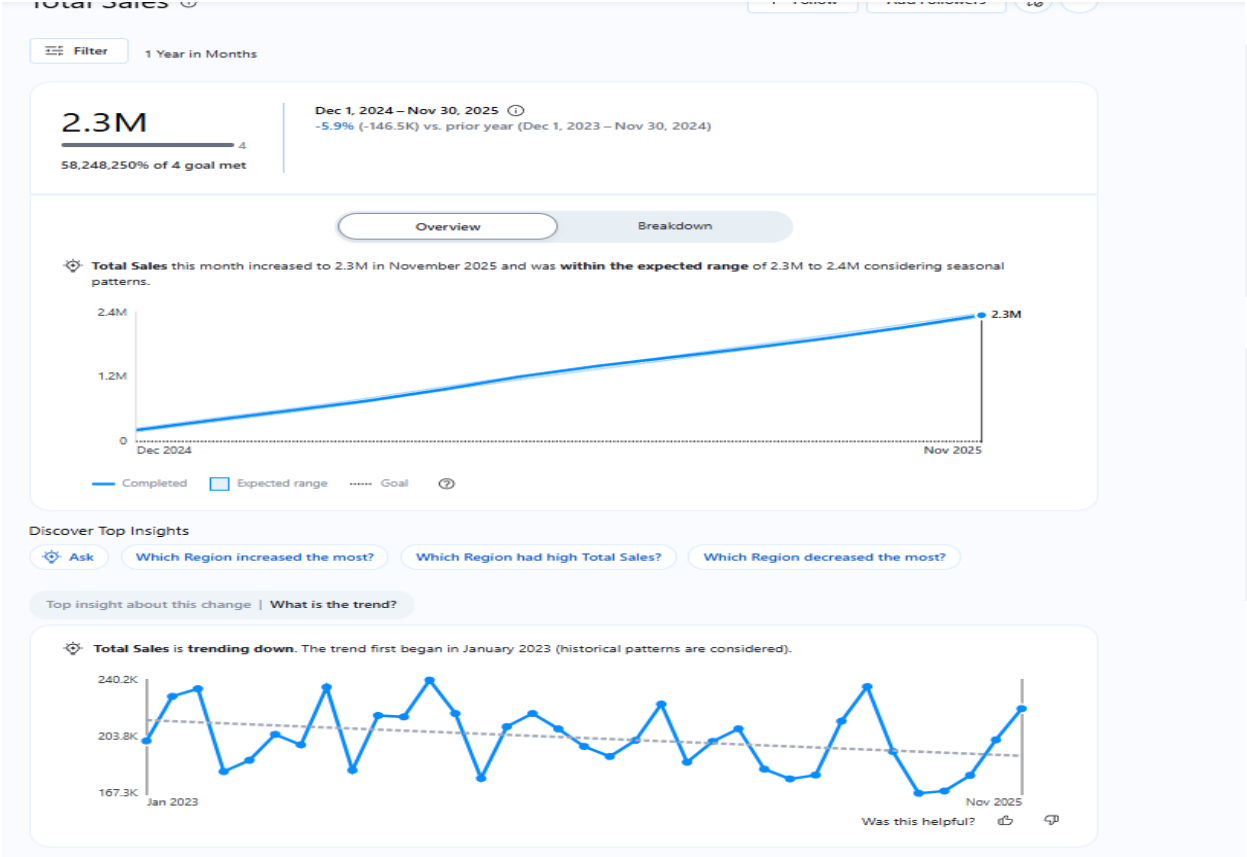
Tableau Pulse on Tableau Cloud

Metric

Sum of Sales, monthly,
compared against prior year

The insight under analysis

Section 1 · two statements, one screen



"...within the expected range... considering seasonal patterns"

Reassuring.

"Total Sales is trending down... since January 2023"

Concerning.

What the AI did well

Section 2 · credit before critique

1	**Accurate**	The corrected figure, –5.9%, matched the source data.
2	**Unrequested**	It found the trend and dated it to January 2023, unprompted.
3	**Relevant**	Its follow-up questions were analytically sensible.
4	**Honest about gaps**	Missing data was flagged in plain language.

What the AI got wrong

Section 2

−65.6%

what Pulse reported

−5.9%

the true figure

Incomparable periods

Four months compared against twelve. No caveat.

Absence drawn as zero

A flat line at zero beneath “no data available”.

No plausibility check

“20,151,400% of goal met” passed unchallenged.

(Szafir, 2018)

Bias analysis

Section 3 · five sources of bias in one output

DATASET	36 synthetic monthly points. No pricing, promotions or market context.
SAMPLING	Three observations per calendar month cannot establish a season.
ALGORITHMIC	**The model assumes seasonality exists — it presumed a pattern, then fitted a range around it.**
SELECTION	Measure, granularity and comparison were my choices, made before any analysis ran.
CONFIRMATION	“Sales are declining” is a story a manager may already half-believe.

(Gebru et al., 2021; Pandey et al., 2014)

Ethical considerations

Section 4

Transparency

The seasonal model behind “expected range” is never shown.

Explainability

The insight is a conclusion, with no route back to the reasoning.

Accountability

Responsibility for acting on −65.6% would have been mine alone.

Trustworthiness

Accurate and inaccurate figures looked identical.

(Kaur et al., 2020)

Human vs AI interpretation

Section 5

PULSE SAID

Normal for the season

Within an expected range, seasonally adjusted.

THE DATA SHOWS

No stable seasonal shape

2023: low April, high December. 2025: high May, low July.

AGREE the direction is real. Annual sales fall in order: €2.53m, €2.45m, €2.32m.

DISAGREE “expected range for the season” rests on a season the data does not contain.

STILL NEEDED units sold are flat; revenue per unit fell €155 to €147. Pricing and promotion context before any decision.

Recommendations

Section 6

01 **Show the data’s coverage window on the metric itself.**

02 **Never draw absent data as zero - break the line.**

03 **State the model behind any claim, and report confidence.**

04 **Treat automated insights as prompts, not conclusions.**

References

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J.W., Wallach, H., Daumé III, H. and Crawford, K. (2021) 'Datasheets for datasets', Communications of the ACM, 64(12), pp. 86–92.

Jones, B. (2019) Avoiding Data Pitfalls. Hoboken, NJ: John Wiley & Sons.

Kaur, H., Nori, H., Jenkins, S., Caruana, R., Wallach, H. and Wortman Vaughan, J. (2020) 'Interpreting interpretability', in Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. New York: ACM.

Pandey, A.V., Manivannan, A., Nov, O., Satterthwaite, M. and Bertini, E. (2014) 'The persuasive power of data visualization', IEEE Transactions on Visualization and Computer Graphics, 20(12), pp. 2211–2220.

Szafir, D.A. (2018) 'The good, the bad, and the biased', Interactions, 25(4), pp. 26–33.

Tableau Create metrics with Tableau Pulse. Available at: help.tableau.com (Accessed: 6 September 2026).